Critical Ising Chain: One-Point Functions in BCFT and Two-Point Functions in CFT

Mini-project report

2026-07-19

Fibonacci category

1 Setup

We study the critical transverse-field Ising (TFI) chain

$$H = -\sum_i \sigma_i^x - \sum_i \sigma_i^z \sigma_{i+1}^z,$$

whose continuum limit is the Ising CFT with central charge $c = 1/2$. The two relevant primary operators are the spin field σ (scaling dimension $\Delta_\sigma = 1/8$) and the energy field ε ($\Delta_\varepsilon = 1$). On the lattice they correspond to

$$\sigma_i^z \leftrightarrow \sigma, \qquad \sigma_i^x \leftrightarrow c_0 + c_1 \varepsilon,$$

i.e. the transverse spin contains the identity plus the energy operator.

We diagonalize H exactly (sparse Lanczos, `Arpack.eigs`) for chains of length $L = 10, 12, 14, 16$ and compute ground-state expectation values.

2 One-point function: free boundary conditions (BCFT)

2.1 Theory

On the upper half plane with a boundary condition a, the one-point function of a primary is fixed by conformal invariance up to a constant:

$$\langle \varphi(x) \rangle_a = A_a (2y)^{-\Delta},$$

where y is the distance to the boundary. Mapping the half plane to a strip of width L (free boundaries on both sides) gives

$$\langle \varphi(x) \rangle = A \left[\left(\frac{L}{\pi} \right) \sin\left(\pi \frac{x}{L} \right) \right]^{-\Delta}.$$

For the free boundary condition the $\mathbb{Z}_2$ symmetry is preserved, so $\langle \sigma \rangle = 0$; the simplest nonzero one-point function is that of the energy operator ε. On the lattice we therefore measure $\langle \sigma_i^x \rangle$, which contains a bulk constant plus the ε profile:

$$\langle \sigma^x(x) \rangle = \frac{2}{\pi} + A \left[\left(\frac{L}{\pi} \right) \sin\left(\pi \frac{x}{L} \right) \right]^{-\Delta_\varepsilon}.$$

2.2 Results

The exact-diagonalization profile for the open chain is fitted to the formula above with the bulk constant fixed to its known value $c_0 = 2/\pi$, dropping the two outermost sites on each side. Figure 1 shows the data and fit for $L = 16$.

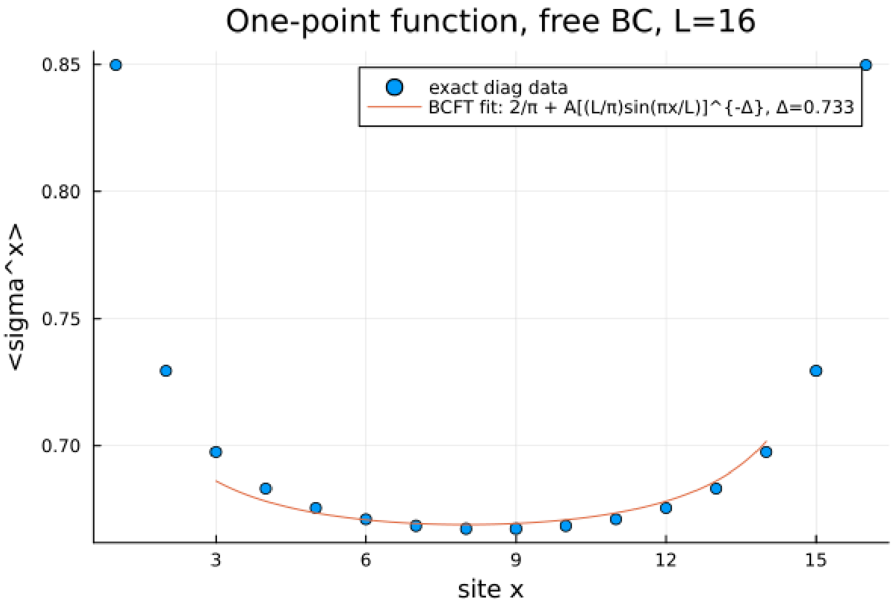

Figure 1: One-point function $\langle \sigma^x(x) \rangle$ for the open (free BC) chain, $L = 16$, with the BCFT fit.

The fitted exponent depends on L through subleading corrections; extrapolating linearly in $1/L$ (Figure 2) gives

$$\Delta_\varepsilon = 1.07,$$

consistent with the BCFT value $\Delta_\varepsilon = 1$.

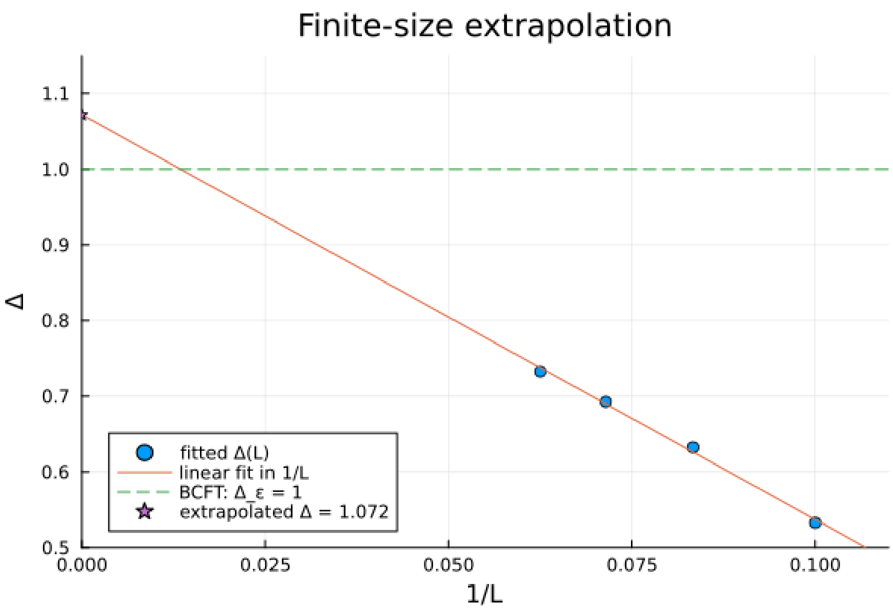

Figure 2: Finite-size extrapolation of the fitted one-point exponent.

3 Two-point functions: periodic boundary conditions (CFT)

3.1 Theory

On a ring of circumference L, translational and conformal invariance fix the two-point function of a primary to

$$\langle \varphi(r)\varphi(0)\rangle = A\left[\left(\frac{L}{\pi}\right)\sin\left(\pi\frac{r}{L}\right)\right]^{-2\Delta},$$

the "chord distance" version of the plane result $r^{-2\Delta}$.

We compute the connected, translation-averaged correlators

$$C_{zz}(r) = \langle \sigma_i^z \sigma_{i+r}^z\rangle, \qquad C_{xx}(r) = \langle \sigma_i^x \sigma_{i+r}^x\rangle - \langle \sigma^x\rangle^2,$$

which probe the spin ($\Delta_\sigma = 1/8$) and energy ($\Delta_\varepsilon = 1$) primaries, respectively.

3.2 Results

Figure 3 and Figure 4 show the correlators for $L = 16$ together with the CFT fits (points at $r = 1$ are excluded from the fit as short-distance lattice artifacts), and the extrapolations of the fitted exponents in $1/L$.

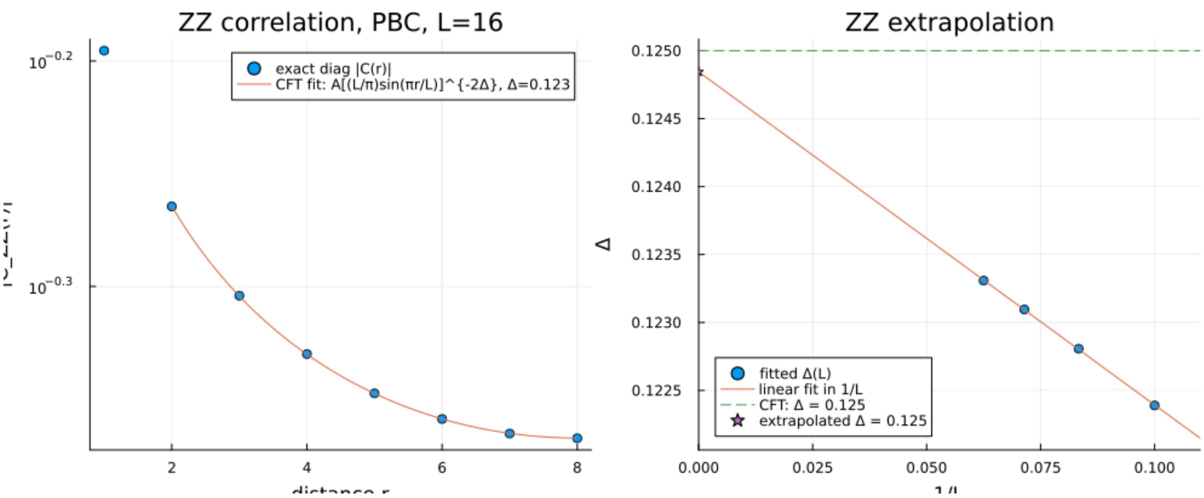

Figure 3: $C_{zz}(r)$ on the ring ($L = 16$) with CFT fit, and finite-size extrapolation of the exponent.

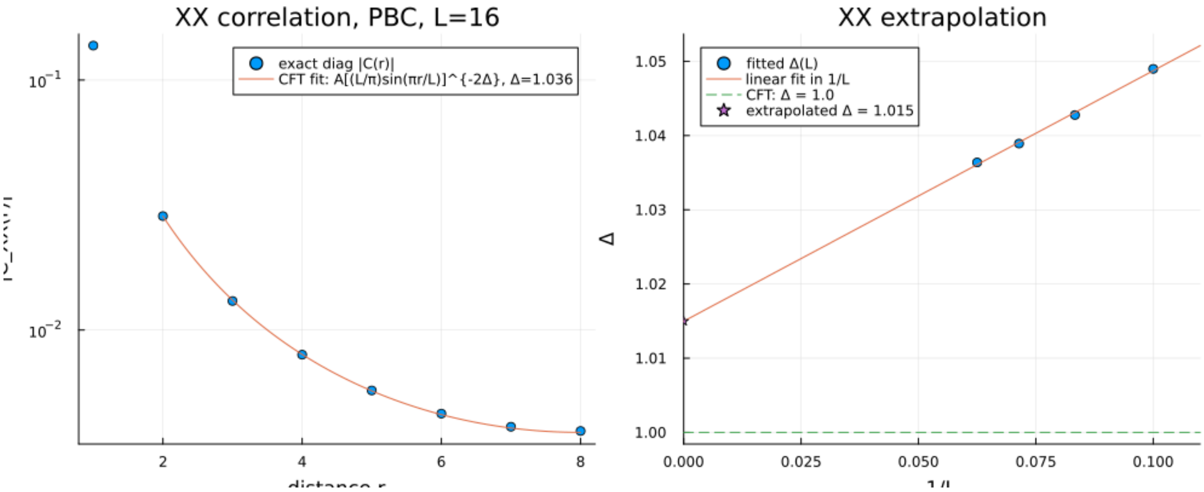

Figure 4: Connected $C_{xx}(r)$ on the ring ($L = 16$) with CFT fit, and finite-size extrapolation of the exponent.

The extrapolated scaling dimensions are

$$\Delta_\sigma = 0.125 \quad \left(\text{exact: } \frac{1}{8}\right), \qquad \Delta_\varepsilon = 1.02 \quad (\text{exact: } 1).$$

4 Summary

Quantity	Observable	Fitted Δ ($L \to \infty$)	Exact Δ
One-point, free BC	σ^x (ε)	1.07	1
Two-point, PBC	$\sigma^z\sigma^z$ (σ)	0.125	1/8
Two-point, PBC	$\sigma^x\sigma^x$ (ε)	1.02	1

Exact diagonalization of the critical TFI chain at modest sizes, combined with the (B)CFT functional forms and a $1/L$ extrapolation, reproduces the Ising CFT scaling dimensions to a few percent. The finite-size corrections are strongest for the boundary one-point fit, where subleading boundary contributions compete with the small one-point amplitude.

Code: `free_BC_corr.jl` (one-point, open chain), `PBC_corr.jl` (two-point, ring).